

BSCE Board Exam Practice Questions

Question 1 (Math)

A steel rod 10 meters long has a cross-sectional area of 0.002 m^2 . If a tensile force of 50,000 N is applied, what is the stress in the rod?

- A. 20,000,000 Pa
- B. 25,000,000 Pa
- C. 30,000,000 Pa
- D. 35,000,000 Pa
- E. 40,000,000 Pa

Correct Answer: B

Question 2 (Mathematic)

A simply supported beam of 6 meters span carries a uniformly distributed load of 12 kN/m. What is the maximum bending moment at midspan?

- A. 48 kN·m
- B. 54 kN·m
- C. 60 kN·m
- D. 72 kN·m
- E. 108 kN·m

Correct Answer: 54 kN·m

Question 3 (Mathematics)

A concrete mix has a water-cement ratio of 0.45. If the weight of cement used is 400 kg, what is the weight of water required?

- A. 160 kg
- B. 175 kg
- C. 180 kg
- D. 190 kg
- E. 200 kg

Correct Answer: C

Question 4 (Mathematics)

A circular column has a diameter of 500 mm and is subjected to an axial compressive load of 1,500 kN. What is the average compressive stress?

- A. 5.86 MPa
- B. 6.43 MPa
- C. 7.07 MPa
- D. 7.64 MPa
- E. 8.12 MPa

Correct Answer: D

Question 5

Which of the following materials has the highest compressive strength?

- A. Timber

- B. Brick masonry
- C. Reinforced concrete
- D. Structural steel
- E. Fiber reinforced polymer

Question 6

Refer to Image 1. Based on the truss diagram shown, what is the force in member BC if the applied load at joint B is 80 kN downward?

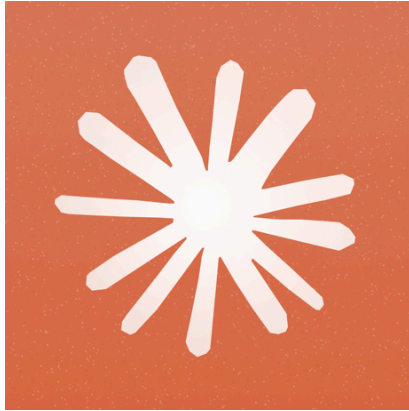


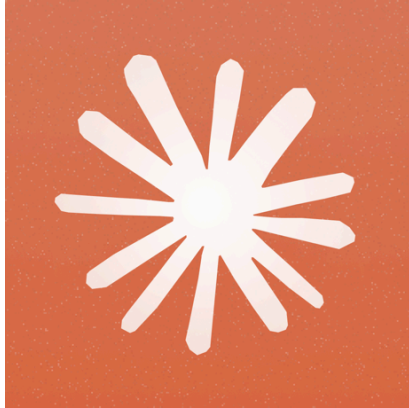
Image 1: A simple Pratt truss with 3 panels, pinned at A, roller at D, with an 80 kN downward load at joint B

- A. 40 kN tension
- B. 60 kN compression
- C. 80 kN tension
- D. 80 kN compression
- E. 100 kN tension

Correct Answer: C

Question 7

Refer to Image 2. Based on the shear force diagram shown, at what location along the beam does the maximum bending moment occur?



- A. At the left support
- B. At 1 meter from the left support
- C. At midspan
- D. At 2 meters from the right support
- E. At the right support

Correct Answer: C

(Image 2: A shear force diagram of a simply supported beam with a point load at midspan, showing positive shear on the left half and negative shear on the right half)